SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Naveen M M (192311166) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Naveen M M

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers.

Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoTenabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

SHORT ANSWER TEST – OSI MODEL CALCULATIONS

1. File Transfer using Transport and Data Link Layers

Given:

- File Size = 150 MB = 150 × 1024 × 1024 = 157,286,400 bytes
- Segment Size = 1500 bytes
- Data Link Overhead = 40 bytes/frame
- Bandwidth = 50 Mbps

Calculation:

Number of Segments

= 157,286,400 ÷ 1500

= 104,857.6 ≈ 104,858 segments

Number of Frames

= 104,858 frames

Total Data Link Overhead

= 104,858 × 40

= 4,194,320 bytes (≈ 4 MB)

Total Data Transmitted

= 157,286,400 + 4,194,320

= 161,480,720 bytes

Convert to bits

= 161,480,720 × 8

= 1,291,845,760 bits

Transmission Time

= 1,291,845,760 ÷ 50,000,000

= 25.84 seconds

Answer:

• Segments = 104,858

• Frames = 104,858

• Total Overhead = 4,194,320 bytes

• Transmission Time = 25.84 seconds

Explanation:

The Transport Layer divides the file into segments, provides sequencing, acknowledgements, and flow control. The Data Link Layer converts segments into frames, adds headers and trailers for MAC addressing and error detection, ensuring reliable communication.

2. Sliding Window Protocol

Given:

• Window Size = 6 frames

• Total Frames = 48

Calculation:

Number of Windows

= 48 ÷ 6

= 8 windows

Retransmissions

= 1 frame lost per window

= 8 retransmissions

Answer:

• Windows Required = 8

• Retransmissions = 8 frames

Explanation:

Flow control prevents the sender from overwhelming the receiver. The sliding window allows multiple frames to be sent before receiving acknowledgements, improving communication efficiency.

3. IP Fragmentation

Given:

- IP Packet = 12,000 bytes
- MTU = 1500 bytes
- IP Header = 20 bytes

Maximum Data per Fragment

= 1500 − 20

= 1480 bytes

Fragments

= 12,000 ÷ 1480

= 8.11 ≈ 9 fragments

Header Overhead

= 9 × 20

= 180 bytes

Answer:

- Fragments = 9
- Header Overhead = 180 bytes

Explanation:

The Network Layer fragments packets larger than the MTU. Each fragment contains its own IP header, and the destination reassembles all fragments to reconstruct the original packet.

4. Sliding Window Protocol

Given:

• Window Size = 6 frames

• Total Frames = 48

Calculation:

Windows

= 48 ÷ 6

= 8

Retransmissions

= 8

Answer:

• Windows Required = 8

• Retransmissions = 8 frames

Explanation:

Flow control regulates the sender’s transmission rate according to the receiver’s capacity, reducing congestion and improving communication efficiency.

5. Session Layer Synchronization

Given:

• Video File = 600 MB

• Checkpoint every 60 MB

• Failure after 390 MB

Calculation:

Completed Checkpoints

= 390 ÷ 60

= 6.5

= 6 checkpoints

Last Checkpoint

= 360 MB

Data to Retransmit

= 390 − 360

= 30 MB

Answer:

- Checkpoints Created = 6
- Data Retransmitted = 30 MB

Explanation:

The Session Layer inserts synchronization checkpoints so communication can resume from the latest checkpoint instead of restarting the entire transmission.

6. Presentation Layer Compression

Given:

- Original File = 900 MB
- Compression = 40%
- Encryption Overhead = 5%

Calculation:

Compressed Size

= 900 × 60%

= 540 MB

Encrypted Size

= 540 × 1.05

= 567 MB

Reduction

= 900 − 567

= 333 MB

Percentage Reduction

= (333 ÷ 900) × 100

= 37%

Answer:

• Compressed Size = 540 MB

• Encrypted Size = 567 MB

• Total Reduction = 37%

Explanation:

The Presentation Layer compresses data to reduce transmission size and encrypts it to provide confidentiality and secure communication.

7. FTP File Transfer

Given:

• File Size = 3 GB = 3072 MB

• Request Size = 3 MB

• Throughput = 120 Mbps

Calculation:

Number of Requests

= 3072 ÷ 3

= 1024 requests

File Size in Megabits

= 3072 × 8

= 24,576 Mb

Transmission Time

= 24,576 ÷ 120

= 204.8 seconds

Answer:

• Requests = 1024

• Transmission Time = 204.8 seconds

Explanation:

The Application Layer provides FTP services, user authentication, file transfer management, and reliable communication between users and network applications.

8. End-to-End Delay

Given:

• Application = 4 ms

• Transport = 3 ms

• Network = 5 ms

• Data Link = 2 ms

• Physical = 1 ms

• Propagation Delay = 18 ms

• Transmission Delay = 12 ms

Calculation:

Processing Delay

= 4 + 3 + 5 + 2 + 1

= 15 ms

Total Delay

= 15 + 18 + 12

= 45 ms

Answer:

• Total End-to-End Delay = 45 ms

Explanation:

Each OSI layer processes the data before transmission. Propagation delay is the travel time through the communication medium, while transmission delay is the time required to send all bits.

9. Protocol Overhead

Given:

• Useful Data = 80 MB = 83,886,080 bytes

• Frame Size = 1600 bytes

• Overhead = 80 bytes/frame

Useful Payload per Frame

= 1600 − 80

= 1520 bytes

Calculation:

Frames Required

= 83,886,080 ÷ 1520

= 55,188.21 ≈ 55,189 frames

Total Overhead

= 55,189 × 80

= 4,415,120 bytes

Useful Data Sent

= 55,189 × 1520

= 83,887,280 bytes

Total Data Sent

= 83,887,280 + 4,415,120

= 88,302,400 bytes

Transmission Efficiency

= (83,887,280 ÷ 88,302,400) × 100

≈ 95%

Answer:

- Frames = 55,189
- Total Overhead = 4,415,120 bytes
- Transmission Efficiency = 95%

Explanation:

Protocol overhead provides addressing, synchronization, and error detection. Although necessary for reliable communication, higher overhead reduces bandwidth efficiency.

10. Healthcare Network Communication

Given:

- Original File = 240 MB
- Compression = 30%
- Segment Size = 1400 bytes
- Data Link Overhead = 60 bytes/frame

Calculation:

Compressed Size

= 240 × 70%

= 168 MB

Compressed Size in Bytes

= 168 × 1024 × 1024

= 176,160,768 bytes

Number of Segments

= 176,160,768 ÷ 1400

= 125,829.12 ≈ 125,830 segments

Protocol Overhead

= 125,830 × 60

= 7,549,800 bytes

Final Data Transmitted

= 176,160,768 + 7,549,800

= 183,710,568 bytes

≈ 175.2 MB

Answer:

- Compressed File Size = 168 MB
- Segments = 125,830
- Protocol Overhead = 7,549,800 bytes
- Final Data Transmitted = 183,710,568 bytes (≈175.2 MB)

Explanation:

The Presentation Layer compresses data for efficient transmission. The Transport Layer divides it into segments and ensures reliable delivery. The Data Link Layer adds headers and trailers for framing and error detection, while the Physical Layer transmits the complete bit stream. Together, these OSI layers provide secure, reliable, and efficient communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, wellexplained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Wellstructured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
		and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand